
SOFTWARE REQUIREMENTS SPECIFICATION(SRS).

CROWDSENSED DRIVE TEST MOBILE APPLICATION FOR COVERAGE PREDICTION AND COVERAGE HOLE ANALYSIS.

Document Property	Value
Document title	Software Requirements Specification (SRS)
Date	10 May 2026
Project	Design & Implementation of a Crowdsensed Drive Test Mobile App
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Table of Contents

1. Introduction and Purpose
 2. Overall System Description
 3. Functional Requirements
 4. Non-Functional Requirements
 5. System Constraints and Assumptions
 6. External Interface Requirements
 7. Stakeholder Validation Checklist
- Appendix A: Glossary of Terms.
- Appendix B: References.
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1. Introduction and Purpose

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for the Crowdsensed Drive Test Mobile Application, a smartphone-based system for collecting cellular network measurements and transmitting them to a backend analytics platform for mobile network coverage prediction and coverage hole analysis.

This document is intended for use by the development team, project supervisor, and all identified stakeholders as the authoritative specification of what the system will do and the quality standards it must meet. It forms the basis for system design, user interface design, and database design.

1.2 Document Scope

This SRS covers:

- The mobile collector application (Android and iOS)
- The backend ingestion and analytics platform
- The web-based analytics dashboard

This SRS does not cover system architecture design, database schema, UI wireframes, or implementation code, those are addressed in Tasks 4, 5, and 6 respectively.

1.3 Intended Audience

Reader	Purpose
Development team	Implementation guidance — what to build and to what standard
Project supervisor	Validation of scope and completeness
Drive test technicians	Verification that operational requirements are correctly captured
RF planning engineers	Verification that dashboard and analysis requirements meet professional needs
Data privacy advocates	Verification that GDPR and privacy requirements are adequately specified
System architects	Input to context diagram, use case, sequence, and class diagram design

2. Overall System Description

2.1 Product Perspective

The Crowdsensed Drive Test Mobile Application is a new system that partially replaces and partially complements traditional professional drive test equipment. Traditional systems use dedicated hardware (TEMS Pocket, Rohde & Schwarz TSME) mounted in specialised vehicles with roof antennas, Single Board Computers, and control laptops. This application leverages the sensing capabilities of commodity smartphones to achieve similar measurement objectives at significantly lower cost and with broader geographic coverage through crowdsensing.

The system operates as a three-tier architecture: mobile collector (edge IoT device), cloud backend (data ingestion and analytics), and web dashboard (visualisation and reporting). The mobile app acts as a Class-2 IoT device, operating autonomously to collect, buffer, and transmit measurements without requiring continuous human supervision.

2.2 Product Functions — Summary

Function	Description	Component
RF measurement collection	Collect RSRP, RSRQ, RSSI, SINR, CQI, cell identity, and network type at 1 Hz	Mobile app
GPS geo-referencing	Stamp every sample with GPS coordinates, accuracy, speed, and timestamp	Mobile app
Offline buffering	Store samples locally when no connectivity; upload automatically on reconnection	Mobile app
Throughput testing	DL/UL speed test with associated RF and GPS context	Mobile app
Call drop detection	Detect and alert on dropped calls during voice session testing	Mobile app
Data transmission	Secure encrypted upload of measurement batches to backend API	Mobile app + backend
Data ingestion	Receive, validate, and store payloads from concurrent devices	Backend
Coverage heatmap	Generate interpolated signal strength map from sample points	Backend + dashboard
Coverage hole detection	Flag areas where RSRP falls below configurable threshold	Backend + dashboard
Reporting and export	CSV and PDF export of session data for engineering analysis	Dashboard
Privacy compliance	Anonymise identifiers; enforce GDPR data handling standards	All components

2.3 User Classes

User Class	Frequency of Use	Technical Level	Primary Interaction
Drive Test Technician	Daily — in-vehicle sessions	Medium — familiar with RF tools	Mobile app collector screen
Network Planning Engineer	Weekly — post-session analysis	High — RF and network expertise	Web dashboard — heatmap and holes
RF Optimisation Engineer	Weekly — trend analysis	High — signal processing expertise	Web dashboard — RSRP/RSRQ trends
NOC Analyst	Daily — alert monitoring	Medium — network operations	Web dashboard — alerts and holes
Executive	Monthly — KPI review	Low — business focus	Web dashboard — summary KPIs
Data Scientist	Periodically — model training	High — ML and data engineering	Backend API — raw data access

2.4 Operating Environment

- Mobile app: Android 10 (API 29) and above; iOS 14 and above; minimum 2 GB RAM; GPS-enabled device; SIM card with active cellular subscription.
 - Deployment context: in moving vehicles (30–120 km/h), outdoor environments, direct sunlight conditions, potential GPS signal degradation in urban canyons.
 - Backend: cloud-hosted (AWS, GCP, or Azure); REST API; PostgreSQL or equivalent relational database with PostGIS spatial extension.
 - Dashboard: modern web browser (Chrome 90+, Firefox 88+, Edge 90+).
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3. Specific Functional Requirements

3.1 Mobile Application- Data Collection

- **FR-01** - Real-time RF measurement collection.

Priority: Must Have (P1).

Rationale: Core application function, without RF measurements the app has no purpose.

Verification: Automated test: initiate session, verify all RF fields populated in sample payload within 2 seconds.

- **FR-02** - GPS geo-referencing of every sample

Priority: Must Have (P1) .

Rationale: Coverage maps require location context for every measurement point.

Verification: Automated test: verify GPS fields (lat, lon, accuracy, speed, bearing, timestamp) present in every sample.

- **FR-03** - Layer 3 (L3) signalling message capture.

Priority: Could Have (P3)

Rationale: Enables advanced signalling analysis for professional drive test use cases.

Verification: Manual test: verify L3 RRC and NAS messages logged on diagnostic-enabled Android device.

- **FR-04** - Voice session testing with Harvard sentences.

Priority: Could Have (P3) .

Rationale: Evaluates speech quality under varying signal, handover, and interference conditions.

Verification: Manual test: initiate voice session, verify MOS calculation, call setup time, and drop event logging.

- **FR-05** - Downlink and uplink throughput speed test.

Priority: Should Have (P2) .

Rationale: Key QoS metric quantifies data performance alongside signal quality.

Verification: Automated test: trigger speed test, verify DL/UL results logged with associated RSRP and GPS coordinates.

- **FR-06** - Call drop detection and operator alert.

Priority: Should Have (P2).

Rationale: Session integrity depends on knowing when calls drop, mirrors the vehicle rig audio alarm.

Verification: Manual test: simulate call drop, verify audible alarm and visual banner alert fire within 2 seconds

3.2 Mobile Application - Data Management

- **FR-07** - Offline local buffering with automatic upload.

Priority: Must Have (P1).

Rationale: Drive tests occur in coverage-weak areas, offline operation is a fundamental field requirement.

Verification: Field test: disable connectivity, collect samples for 90 minutes, re-enable connectivity, verify all samples uploaded with no loss

3.3 Backend Platform - Data Ingestion and Analysis

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- **FR-08** - Multi-device measurement payload ingestion.

Priority: Must Have (P1).

Rationale: Crowdsensing requires simultaneous collection from many devices across a city.

Verification: Load test: simulate 50 concurrent device uploads, verify all payloads stored correctly within SLA.

- **FR-09** - Coverage heatmap with spatial interpolation.

Priority: Should Have (P2).

Rationale: Core visualisation output, enables RF planners to see signal distribution across the drive route.

Verification: Visual inspection: verify heatmap renders correctly with colour-coded RSRP levels on geographic map.

- **FR-10** - Automated coverage hole detection and alerting.

Priority: Must Have (P1).

Rationale: Primary analytical deliverable, identifies locations where signal falls below acceptable threshold

Verification: Automated test: inject sample with RSRP below -110 dBm threshold, verify coverage hole flag and alert generated.

3.4 Dashboard - Reporting

- **FR-11** - CSV and PDF session data export.

Priority: Should Have (P2).

Rationale: RF planners need exportable data for post-session engineering analysis and documentation.

Verification: Manual test: trigger export from dashboard, verify CSV contains all measurement fields, PDF renders correctly

3.5 Privacy and Compliance

- **FR-12** - Device and user identifier anonymisation.

Priority: Must Have (P1).

Rationale: GDPR compliance is a legal obligation given the collection of geo-referenced user data.

Verification: Code review: verify no personally identifiable information present in transmitted payloads; penetration test confirms anonymisation.

4. Non-Functional Requirements

4.1 Performance Requirements

ID	Requirement	Metric	Test Method
NFR-01	Measurement sampling rate	≥1 sample per second during active session	Automated timing test over 60-second window
NFR-05	Backend API ingestion response time	≤500 ms per payload request under normal load (≤50 concurrent)	API load test with simulated payloads
NFR-06	Dashboard coverage map refresh latency	≤5 seconds from upload completion to map update	End-to-end timing test from upload trigger to map render

4.2 Availability and Reliability Requirements

ID	Requirement	Metric	Test Method
NFR-02	Offline session operation without data loss	≥90 minutes continuous operation with no connectivity	Field test: disable network, run full session, verify zero sample loss
NFR-09	Concurrent device support	≥50 simultaneous device uploads without performance degradation	Cloud load test: 50 concurrent upload clients, verify SLA compliance

4.3 Security Requirements

ID	Requirement	Standard	Test Method
NFR-07	Data in transit encryption	TLS 1.3 minimum for all API communication	Penetration test; network traffic analysis
NFR-07	Data at rest encryption for local buffer	AES-256 for locally stored measurement samples	Code review of local storage implementation
FR-12	GDPR compliance	No PII in transmitted data; user consent flow; data retention policy	Privacy audit; DPA (Data Protection Authority) checklist

4.4 Usability Requirements

ID	Requirement	Metric	Test Method
NFR-08	One-handed operation on target device size	Fully operable on screens 5–7 inches with one hand	Usability test: target users complete core tasks one-handed
NFR-08	Sunlight readability	Minimum contrast ratio 4.5:1 per WCAG 2.1 Level AA	Automated contrast checker; field test in direct sunlight
NFR-04	Application installed size	≤50 MB total installed size	Build artifact size measurement

4.5 Power and Resource Requirements

ID	Requirement	Metric	Test Method
NFR-03	Battery consumption during active session	≤15% battery per hour during continuous measurement collection	Measure battery level at session start and end over 1-hour test
NFR-04	Local storage consumption during offline session	Maximum 500 MB for a 90-minute 1Hz session buffer	Calculate: 1 sample/s × 90 min × estimated payload size

4.6 Compatibility Requirements

ID	Requirement	Scope	Rationale
NFR-10	Android platform support	Android 10 (API level 29) and above	TelephonyManager APIs for RSRP/RSRQ require API 29 minimum
NFR-10	iOS platform support	iOS 14 and above	Covers majority of active iPhone devices; GPS and speed APIs available
NFR-10	Backend API	RESTful JSON over HTTPS; OpenAPI 3.0 specification	Standard interoperability; enables third-party integrations
NFR-10	Dashboard browser support	Chrome 90+, Firefox 88+, Edge 90+	Covers 95%+ of enterprise browser market share

5. System Constraints and Assumptions

5.1 Technical Constraints

- iOS platform constraint: Apple does not expose RSRP, RSRQ, PCI, Cell ID, or EARFCN to third-party applications via public APIs. The iOS build will be limited to network type (2G/3G/4G/5G indicator), throughput, latency, GPS, and device metadata. Full RF measurement capability requires Android. This is a hard platform constraint that cannot be resolved without Apple entitlements or a carrier agreement.
- L3 message constraint: Layer 3 RRC/NAS message access on Android requires either root access or a manufacturer-specific diagnostic port agreement. Standard consumer smartphones do not expose L3 messages via TelephonyManager. This requirement (FR-03) is therefore deferred to Could Have and requires resolution with device vendors before implementation.
- GPS accuracy constraint: GPS positioning accuracy degrades in urban canyons, under dense tree cover, and in tunnels. Samples with GPS accuracy readings above 50 metres should be flagged with a quality indicator in the data payload rather than discarded, as even imprecise location data contributes to coverage models when properly weighted.
- Coverage prediction algorithm: The specific interpolation or machine learning algorithm to be used for coverage prediction has not been selected. This is a critical missing requirement (M-4) that must be resolved before backend development can begin.

5.2 Regulatory Constraints

- GDPR (EU) 2016/679: The application collects continuous geo-referenced location data. Users must provide explicit informed consent before data collection begins. Data must be anonymised before transmission. A data retention policy must be defined. Users must have the right to request deletion of their data.
- Local telecommunications regulations: In some jurisdictions, collecting and transmitting radio frequency measurements may require notification to or approval from the national telecommunications authority. The deployment team must verify applicable regulations in each target country.

5.3 Assumptions

- Target devices have a minimum of 2 GB RAM, 16 GB available storage, a functioning GPS receiver, and an active cellular SIM card with data connectivity.
 - The backend platform will be deployed on a cloud provider with auto-scaling capabilities to handle variable concurrent device loads.
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- Users will have granted location (precise), phone state, and storage permissions before the application initiates a measurement session.
 - Network engineers reviewing the dashboard have basic GIS literacy sufficient to interpret coverage heatmaps.
 - The application will be distributed through official app stores (Google Play Store and Apple App Store) or enterprise MDM deployment for professional users.

6. External Interface Requirements

6.1 User Interfaces

Screen	Primary Users	Key Elements Required
Session setup screen	Drive test technician	Operator ID input; session type selector (voice/data/both); start session button; GPS fix indicator; network type indicator
Live metric screen	Drive test technician	RSRP (primary, top-left), RSRQ, RSSI, SINR displayed prominently; signal bars; network type badge; GPS fix indicator; session timer; stop/upload button; call drop alert banner
Transmission log	Drive test technician	Upload progress; sample count; buffer status; last successful upload timestamp
Coverage heatmap	RF planner, NOC	Interactive map with RSRP colour gradient; coverage hole markers; session filter; date range filter; zoom and pan
Coverage hole list	RF planner, NOC	Sortable table of flagged locations; RSRP value; GPS coordinates; timestamp; severity indicator
Session export	RF planner	Session selector; export as CSV button; export as PDF button; preview panel
KPI dashboard	Executive	Coverage percentage gauge; average RSRP trend; number of holes detected; active sessions count

6.2 Hardware Interfaces

- Cellular modem, accessed via Android TelephonyManager (API 29+) for RF measurements; iOS CoreTelephony for network type only
- GPS receiver, accessed via Android LocationManager or iOS CoreLocation; minimum accuracy 50 metres
- Microphone, required for voice session testing (FR-04, P3); accessed via Android AudioRecord or iOS AVAudioSession

6.3 Software Interfaces

Interface	Protocol	Format	Direction
Mobile app → Backend API	HTTPS REST	JSON payload	Mobile to server
Backend → Dashboard	HTTPS REST / WebSocket	JSON	Server to browser
Backend → Database	PostgreSQL wire protocol	SQL queries	Server to database

Interface	Protocol	Format	Direction
Backend → Map tiles	HTTPS	OpenStreetMap / Mapbox tiles	Server to browser
Dashboard → Export	HTTP download	CSV / PDF	Server to browser

6.4 Communication Interfaces

- Primary uplink: HTTPS REST API over cellular data (4G/5G preferred; 3G minimum for upload)
- Offline buffer: Local SQLite database on the device; flushed to backend on reconnection
- Authentication: OAuth 2.0 device-flow or API key per registered device; JWT tokens for session management

7. Stakeholder Validation Checklist.

This checklist documents the validation status of all requirements. Items marked Outstanding must be resolved with the relevant stakeholder before this SRS can be formally baselined and development commenced.

#	Validation Item	Stakeholder	Status	Action
1	Drive test operational workflow accuracy confirmed	Drive Test Technician	Confirmed	None required
2	Offline 90-min session duration validated against storage	Technician / Engineer	Partial	Calculate: $1\text{Hz} \times 90\text{min} \times \text{payload size} \approx \text{confirm} \leq 500\text{MB}$
3	Secondary research accepted as substitute for MNO interviews	Supervisor	Partial	Formal supervisor sign-off required
4	All 8 stakeholder roles reviewed and requirements approved	All stakeholders	Partial	Formal review session required
5	Dashboard KPI list confirmed by RF planners	Network Planning Engineer	Outstanding	Schedule stakeholder review session
6	Coverage prediction algorithm selected and documented	Data Scientist	Outstanding	Select: Kriging / IDW / ML — document justification
7	Data anonymisation approach reviewed by privacy advocate	Data Privacy Advocate	Outstanding	Draft anonymisation policy; submit for review
8	Alert thresholds and notification channels confirmed	NOC Team	Outstanding	Define RSRP hole threshold; confirm notification method
9	5G NR scope confirmed in or out of v1	Project Team	Outstanding	Team decision and documentation required
10	L3 capture feasibility confirmed with device vendor	Technical Team	Outstanding	Contact Android device vendor for diagnostic API access
11	iOS limitations documented and iOS scope adjusted	Technical Team	Outstanding	Document iOS feature reduction formally in SRS Section 5
12	SIP message types defined and VoLTE scope confirmed	RF / Technical Team	Outstanding	Confirm VoLTE in or out of scope; define SIP methods if in
13	All ambiguities A-1 through A-4 resolved	Project Team + Supervisor	Outstanding	Resolution meeting; document decisions in SRS
14	SRS formally baselined and version-controlled	Project Team	Outstanding	Commit to version control after all sign-offs complete

Appendix A: Glossary of Terms

Term	Definition
RSRP	Reference Signal Received Power — primary LTE/5G signal strength indicator
RSRQ	Reference Signal Received Quality — quality relative to interference and noise
RSSI	Received Signal Strength Indicator — total received wideband power
SINR	Signal to Interference plus Noise Ratio
CQI	Channel Quality Indicator — UE feedback to base station for scheduling
RSCP	Received Signal Code Power — 3G WCDMA equivalent of RSRP
Ec/No	Chip Energy to Noise Ratio — 3G signal quality
EARFCN	E-UTRA Absolute Radio Frequency Channel Number — LTE carrier frequency identifier
NR-ARFCN	New Radio Absolute Radio Frequency Channel Number — 5G NR equivalent of EARFCN
PCI	Physical Cell Identity — unique cell identifier within a network area
MCC	Mobile Country Code — identifies the country of the serving operator
MNC	Mobile Network Code — identifies the operator within a country
TAC	Tracking Area Code — LTE equivalent of Location Area Code
L3	Layer 3 — signalling layer including RRC (Radio Resource Control) and NAS (Non-Access Stratum) messages
MOS	Mean Opinion Score — perceptual voice quality rating used in ITU-T P.800
SBC	Single Board Computer — embedded controller used in professional drive test vehicle rigs
RAT	Radio Access Technology — the generation of mobile network in use
MNO	Mobile Network Operator — the company operating the cellular network under test
IoT	Internet of Things — the smartphone functions as a Class-2 IoT sensing device
GDPR	General Data Protection Regulation — EU Regulation 2016/679 on data protection
TLS	Transport Layer Security — cryptographic protocol for secure data transmission
AES	Advanced Encryption Standard — symmetric encryption for data at rest
WGS84	World Geodetic System 1984 — global coordinate reference system used by GPS

Appendix B: References

- IEEE Std 830-1998 — IEEE Recommended Practice for Software Requirements Specifications. IEEE Computer Society.
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